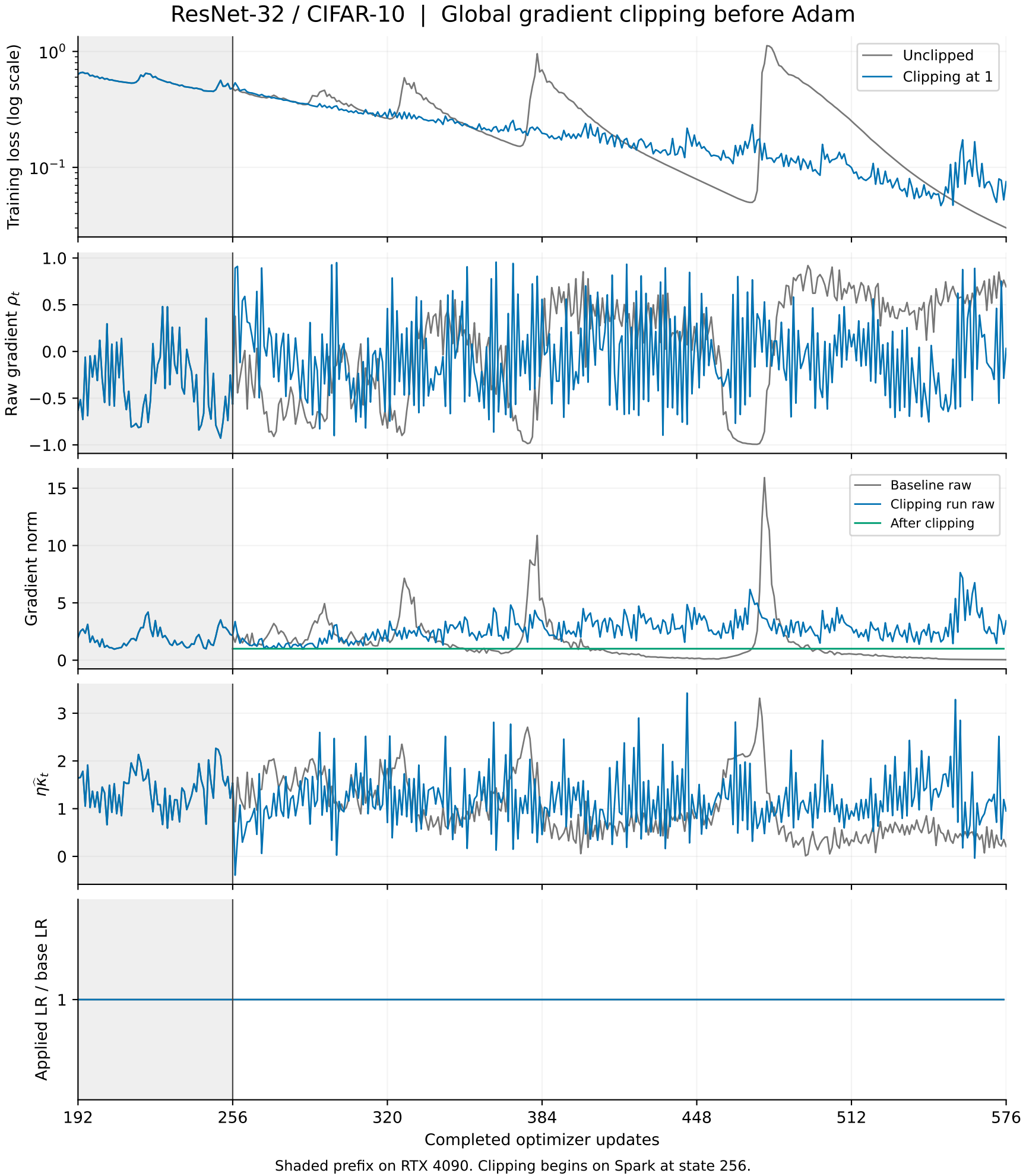
ResNet-32 / CIFAR-10 | Global gradient clipping before Adam
Training loss (log scale)
Unclipped
Clipping at 1
Raw gradient ρ_t
Gradient norm
Baseline raw
Clipping run raw
After clipping
$\hat{\eta\kappa}_t$
Applied LR / base LR
Completed optimizer updates
Shaded prefix on RTX 4090. Clipping begins on Spark at state 256.
Full 50,000-image objective, seed 0, constant LR 0.003.